

Short Exam N°2- Computer Vision 2023

1 h – No document allowed except lecture slides and notes (18 points)

Part 1: Camera calibration.

The camera shown in Figure 1 has its x , y and z axes aligned with the world's y , z and x axes respectively. The world frame's origin is at $(0; -h; 4h)$ in the camera's frame.

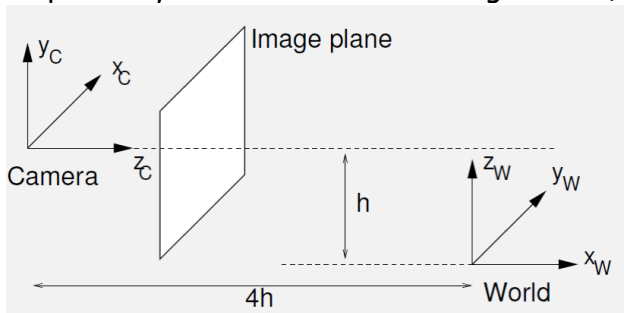


Figure 1

Question 1 (2 points - 5 mn): By deriving a succession of Euclidean transformations, find the camera's extrinsic camera calibration matrix $[R|t]$, such that:

$$\mathbf{X}_C = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \mathbf{X}_W$$

Ans: rotation R_x of 90° around X then a rotation R_z of -90° around Z (so $R = R_x * R_z$), next a translation.

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \quad R_z = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{X}_C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -h \\ 1 & 0 & 0 & 4h \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathbf{X}_W$$

Then

Question 2 (2 points - 5 mn): The "ideal" image plane is placed at $z_c = 1$. (Ideal means that the intrinsic camera matrix K is just a 3×3 identity matrix.) Derive the image coordinates of the vanishing point of the family of lines parallel to the following line, expressed parametrically as:

$$(X_w; Y_w; Z_w) = (2 + 4t; 3 + 2t; 4 + 3t)$$

Ans: $K [R|t] = [R|t]$

Either project then let $t \rightarrow \infty$ or vice versa. You should find

$$\mathbf{x} \stackrel{P}{=} \begin{bmatrix} 1/2 \\ 3/4 \\ 1 \end{bmatrix}$$

$$X_w/Z_w = (2+4t)/(4+3t) = \frac{1}{2} \text{ when } t \rightarrow \infty$$

$$Y_w/Z_w = (3+2t)/(4+3t) = \frac{3}{4} \text{ when } t \rightarrow \infty$$

Question 3 (2 points - 5 mn): The intrinsic calibration matrix maps ideal image positions onto actual positions in pixels. Explain, with the aid of diagrams, the physical meaning of the focal length f , the aspect ratio γ , and the principal point $(u_0; v_0)$, and show why and where they appear in the calibration matrix. (Assume the skew s is negligible.)

Ans : see lecture notes (slides 5 and 21 of lecture 3b)

Question 4 (2 points - 5 mn): The actual camera has $f = 800$ pixels, $\gamma = 0.9$, $s = 0$, and $(u_0; v_0) = (350; 250)$ pixels. Derive the coordinates in the actual image plane of the vanishing point of question 2.

Ans : $[x, y] = [750, 790]$

Part 2: Camera calibration.

Question 5 (2 points - 5 mn): Starting with the projection equation $\mathbf{x} = \mathbf{P}\mathbf{X} = \mathbf{K}[\mathbf{R}|\mathbf{t}]\mathbf{X}$ (all up to scale), describe an algorithm which uses the measured image positions $(x_i; y_i)$ of at least six points with known positions in the world $(X_i; Y_i; Z_i)$ to recover the extrinsic and intrinsic calibration parameters.

Ans : see lecture notes (slide 18 of lecture 4); $\mathbf{P}$ has 12 unknowns, so to recover the extrinsic and intrinsic parameters we need 12 equations, and as we have 2 equations per pair of corresponding points we need 6 points with known positions.

Question 6 (4 points - 15 mn): Let us suppose that $\mathbf{K}$ is defined as below, and that we know the image positions $(x_i; y_i)$ of six points of a unit cube defined by the following positions in the world $(X_i; Y_i; Z_i)$ explain the process to get the coefficients of the matrix $\mathbf{P}$ (up to scale).

$$\mathbf{K} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(x_1; y_1) = (2; 1/2), \quad (X_1; Y_1; Z_1) = (0; 0; 0)$$

$$(x_2; y_2) = (9/4; 1/2), \quad (X_2; Y_2; Z_2) = (0; 1; 0)$$

$$(x_3; y_3) = (9/4; 1), \quad (X_3; Y_3; Z_3) = (0; 1; 1)$$

$$(x_4; y_4) = (2; 1), \quad (X_4; Y_4; Z_4) = (0; 0; 1)$$

$$(x_5; y_5) = (2; 1), \quad (X_5; Y_5; Z_5) = (1; 0; 1)$$

$$(x_6; y_6) = (11/5; 1), \quad (X_6; Y_6; Z_6) = (1; 1; 1)$$

Hint : From the 6 points we get 12 equations, from these equations we get the $\mathbf{R}$ and $\mathbf{T}$ unknowns. Remind that:

$$KR = \begin{bmatrix} (fR_{11} + sR_{21} + u_0R_{31}) & (fR_{12} + sR_{22} + u_0R_{32}) & (fR_{13} + sR_{23} + u_0R_{33}) \\ (\gamma fR_{21} + v_0R_{31}) & (\gamma fR_{22} + v_0R_{32}) & (\gamma fR_{23} + v_0R_{33}) \\ K_{33}R_{31} & K_{33}R_{32} & K_{33}R_{33} \end{bmatrix}$$

Projection matrix is (up to scale)

$$\begin{bmatrix} 2 & 1 & 0 & 8 \\ 1 & 0 & 2 & 2 \\ 1 & 0 & 0 & 4 \end{bmatrix}$$

Part 3: Short questions.

Question 7 (1 point - 5 mn): The intrinsic parameters of two cameras are given by the following matrixes, which of these two cameras would you prefer to use? Explain your choice.

$$K_1 = \begin{bmatrix} 0.6 & 0.3 & -1.4 \\ 0 & 0.4 & 0.9 \\ 0 & 0 & 1 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 0.4 & 0.3 & 0.9 \\ 0 & 0.6 & 1.4 \\ 0 & 0 & 1 \end{bmatrix}$$

Ans: see slide 21 of lecture 3b. We prefer to have positive translation shifts in the image plane (ie k2) than any negative shift (as with k1).

Question 8 (2 points - 10 mn): A pinhole camera projects a 3D point P, located at coordinate $[1; 0; 0]^T$ in the world coordinate system, onto the image at point p using the projection matrix M, as $p = MP$, where:

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Compute the image coordinates of p. Compute the following camera parameters: Focus; Skew; Offset along x axis; Offset along y axis; Scaling.

Ans: see slide 21 of lecture 3b. The first matrix corresponds to K, so:

$$f \text{ sx} = 1$$

$$f \text{ sy} = 1$$

$$S\theta = 0$$

$$ox = 0$$

$$oy = 1$$

$$f \text{ sx} = f \text{ sy}, \text{ so } sx = sy$$

$$p = (-1, -1, -1)^T$$

As the third coeff should be = 1 (with homogeneous coord) we must factorize the value -1, so $p = (1, 1)^T$

Question 9 (1 point - 5 mn): From the specification of M, and the standard definition of the 3D rotation matrix

$$R = \begin{bmatrix} \cos\theta \cos\psi & -\cos\phi \sin\psi + \sin\phi \sin\theta \cos\psi & \sin\phi \sin\psi + \cos\phi \sin\theta \cos\psi \\ \cos\theta \sin\psi & \cos\theta \cos\psi + \sin\phi \sin\theta \sin\psi & -\sin\phi \cos\psi + \cos\phi \sin\theta \sin\psi \\ -\sin\theta & \sin\phi \cos\theta & \cos\phi \cos\theta \end{bmatrix},$$

the transform that maps the world coordinate system of P to the camera coordinate system is characterized by the following rotation angle: $\theta = 90^\circ$, $\phi = 0^\circ$, and $\psi = 0^\circ$. Compute the Euclidean distance between P and the camera center at the moment of capturing the image.

Ans:

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

The rotation of the coordinate system does not impact the computation of the Euclidean distance, if there is no translation then the distance is equal to zero.